

Desplegar un recurso real y medir su coste en Azure

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1. Estimar el coste antes de desplegar

Para esta estimación necesitas estos tres componentes:

- Virtual Machines
- Managed Disks
- IP Addresses

A tener en cuenta: la calculadora está en euros (€) y Azure al tenerlo en Inglés está en dolares (\$).

Configuramos la máquina con los valores indicados.

Virtual Machines: Mi máquina virtual

1 B2ls v2 (2 vCPUs, 4 GB RAM) x 72 Hours (Pay ...)

Upfront: €0.00

Monthly: €2.82

Annual: €33.83

Mi máquina virtual

Give feedback

Get €200 credit plus free monthly amounts of popular services for 12 months—including Virtual Machines. See free amounts

Region:Spain Central

Operating system:Linux

Type:Ubuntu

Tier:Standard

Category:All

Instance Series:All

INSTANCE: (Need help finding the right VM?)B2ls v2: 2 vCPUs, 4 GB RAM, 0 GB Temporary storage, €0.039/hour

1Virtual machines

72Hours

Savings Options

Explore pricing models to help optimize your Azure costs. Learn more

Compute (B2ls v2)

Pay as you go

Savings plan for compute

1 year savings plan (~24% discount)

3 year savings plan (~47% discount)

Reservations

1 year reserved (~41% discount)

3 year reserved (~62% discount)

€2.82Average per month(€0.00 charged upfront)

€2.82Average per month(€0.00 charged upfront)

Managed Disks

€0.00

Storage transactions

€0.00

Bandwidth

€0.00

Upfront cost

€0.00

Monthly cost

€2.82

Annual cost

€33.83

Configuramos el disco con los valores indicados.

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Managed Disks: Mi disco

Managed Disks, Standard SSD, LRS Redundancy...

Upfront: €0.00

Monthly: €2.06

Annual: €24.73

Mi disco

Give feedback

Get €200 credit plus free monthly amounts of popular services for 12 months—including Managed Disks, Standard HDD tier. See free amounts

Region:

Spain Central

Tier:

Standard SSD

Redundancy:

LRS

Disk size:

E4: 32 GiB, 500 Provisioned IOPS, 100 Provisioned MB/s, €2.061/month

Number of Disks

1

Disks

×

€2.06

Per month

=

€2.06

Storage transactions

0

Transaction units (10,000 transactions)

×

€0.0017

Per unit

=

€0.00

An IO unit size of 256 KiB is used to calculate the number of billable transaction units – see this doc for more information.

Add Snapshot

Enable Confidential OS Encryption

Upfront cost

€0.00

Monthly cost

€2.06

Annual cost

€24.73

Configuramos la IP pública.

IP Addresses: Mi dirección IP

Standard (ARM), 1 Static IP Addresses X 72 Hou...

Upfront: €0.00

Monthly: €0.31

Annual: €3.71

Mi dirección IP

Give feedback

Get €200 credit plus free monthly amounts of popular services for 12 months—including Public IP Addresses, Basic (Classic). See free amounts

Region:

Spain Central

Type:

Standard (ARM)

NOTE:

Common price for load balanced or virtual machines. Standard (ARM) public IP preview rates are in effect during this period

Static IP Addresses

1

Addresses

×

72

Hours

=

€0.31

Public IP Prefixes

0

Addresses

×

730

Hours

=

€0.00

Upfront cost

€0.00

Monthly cost

€0.31

Annual cost

€3.71

Cálculo de los tres escenarios:

Escenario	Horas de cómputo	Coste cómputo (mensual)	Coste disco	Coste IP	Total estimado
A. VM encendida todo el laboratorio	72	2,82€	0,20€	0,31€	3,33€
B. VM con apagado automático nocturno (≈ 12 h/día)	36	1,41€	0,20€	0,31€	1,92€
C. VM encendida solo 8 h en total	8	0,31€	0,20€	0,31€	0,82€

Creación del Resource Group

En Azure accedemos a **Resource groups** y creamos el grupo de recursos con la siguiente información:

Create a resource group ...

Basics

Tags

Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ

Azure for Students

Resource group name * ⓘ

rg-practica04-vm-ChristianSalguero

Region * ⓘ

(Europe) West Europe

Agregamos sus Tags:

Basics

Tags

Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name		Value	Resource	
Project	:	Practica04	Resource group	
Department	:	Data	Resource group	
Environment	:	Lab	Resource group	
Owner	:	ChristianSalguero	Resource group	
CostCenter	:	DataNova-Analytics	Resource group	
	:		Resource group	

Cambiamos el scope para crear el Budget correctamente.



Cost Management: rg-practica04-vm-ChristianSalguer...

Resource group

 Overview

 Change scope

 Access control

 Diagnose and solve problems

 Reporting + analytics

-  Cost analysis
-  Exports

 Monitoring

-  Cost alerts
-  Alert rules
-  **Budgets**

Scope :  rg-practica04-vm-ChristianSalguero

Name	↑↓	Scope	↑↓	Reset period
You do not have any budgets.				

Creamos el Budget con la información necesaria.

Create budget ...

Budget

Budget scoping

The budget you create will be assigned to the selected scope. Use additional filters like resource groups to have your budget monitor with more granularity as needed.

Scope

 rg-practica04-vm-ChristianSalguero

Filters

+  Add filter

Budget Details

Give your budget a unique name. Select the time window it analyzes during each evaluation period, its expiration date and the amount.

*Name

Budget-Practica04-VM-ChristianSalguero 

*Reset period ⓘ

Monthly 

*Creation date ⓘ

2026 

September 

1 

*Expiration date ⓘ

2028 

August 

31 

Budget Amount

Give your budget amount threshold

Amount *

2 

Configuramos las alertas.

Create budget ...

Budget

✓ Create a budget ✓ Set alerts

Configure alert conditions and send email notifications based on your spend.

* Alert conditions

Type	% of budget	Amount	Action group
Actual cost ▾	50 ✓	1	None ▾
Actual cost ▾	80 ✓	1.6	None ▾
Actual cost ▾	100 ✓	2	None ▾
Forecasted cost ▾	100 ✓	2	None ▾
Select type ▾	Enter %	-	None ▾

[Manage action group](#) ⓘ

* Alert recipients (email)

Alert recipients (email)

christian.salguero@tajamar365.com ✓	🗑️
example@email.com	

It is recommended to add azure-noreply@microsoft.com to your email allow list to ensure alert mails do not go to your spam folder.

Delegar la Máquina Virtual

Definimos los primeros valores de la máquina virtual. Como entre nuestras regiones permitidas no está **West Europe** he buscado cuál tenía disponible la instancia deseada y esa era **Spain Central**.

Subscription * ⓘ

Azure for Students



Resource group * ⓘ

rg-practica04-vm-ChristianSalguero

[Create new](#)**Instance details**

Virtual machine name * ⓘ

vm-practica04-ChristianSalguero

Region * ⓘ

(Europe) Spain Central

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

No infrastructure redundancy required



Security type ⓘ

Standard



Image * ⓘ

 Ubuntu Server 24.04 LTS - x64 Gen2[See all images](#) | [Configure VM generation](#)

VM architecture * ⓘ



Arm64



x64

Size * ⓘ

Standard_B2ls_v2 - 2 vcpus, 4 GiB memory (\$33.29/month)

[See all sizes](#) | [Customize cores](#)

Run with Azure Spot discount ⓘ

☐

Comparamos la instancia entre la calculadora y Azure y vemos que son la misma con mismo valor (ambas están en dolares \$).

Category:

All

Instance Series:

All

INSTANCE: [\(Need help finding the right VM?\)](#)

B2Is v2: 2 vCPUs, 4 GB RAM, 0 GB Temporary storage, \$0.046/hour

1

×

1

Month

Virtual machines

Savings Options

Explore pricing models to help optimize your Azure costs.

Learn more

Compute (B2Is v2)

☒ Pay as you go

Savings plan for compute ⓘ

- ☐ 1 year savings plan (~24% discount)
- ☐ 3 year savings plan (~47% discount)

Reservations ⓘ

- ☐ 1 year reserved (~41% discount)
- ☐ 3 year reserved (~62% discount)

\$33.29

Average per month

(\$0.00 charged upfront)

= \$33.29

Average per month

(\$0.00 charged upfront)

Definimos los valores del disco.

Administrator account

Authentication type * ⓘ

☒ SSH public key

☐ Password

ⓘ Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username * ⓘ

azureuser

SSH public key source * ⓘ

Generate new key pair

SSH Key Type * ⓘ

☒ RSA SSH Format

☐ Ed25519 SSH Format

Key pair name * ⓘ

vm-practica04-ChristianSalguero_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

Select inbound ports

Select one or more ports

ⓘ All traffic from the internet will be blocked by default. You will be able to

Definimos los valores de red.

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ⓘ

☐

i Encryption at host is not registered for the selected subscription.
[Learn more](#) ↗

OS disk

OS disk size ⓘ

Image default (30 GiB) ▾

OS disk type * ⓘ

Standard SSD ▾

i The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ⓘ

☒

Enable Ultra Disk compatibility ⓘ

☐

i Ultra disk is only supported in Availability Zone(s) 1,2 for the selected VM size Standard_B2ls_v2.

Key management ⓘ

Platform-managed key ▾

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

Public IP ⓘ

(new) vm-practica04-ChristianSalguero-ip

[Create new](#)

Public IP addresses have a nominal charge. [Learn more](#)

NIC network security group ⓘ

☐ None

☒ Basic

☐ Advanced

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

Select inbound ports

Select one or more ports

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Delete public IP and NIC when VM is deleted ⓘ



Enable accelerated networking ⓘ



This size requires accelerated networking

No podemos marcar la opción de **Auto ShutDown** ya que no está disponible en **Spain Central**. Más adelante intento volver a activar esta opción y me da error.

Auto-shutdown

Enable auto-shutdown ⓘ



The selected region does not support auto-shutdown.

Agregamos los Tags a la máquina virtual:

Tags

Name ⓘ	Value ⓘ	
CostCenter	: DataNova-Analytics	
Department	: Data	
Environment	: Lab	
Owner	: ChristianSalguero	
Project	: Practica04	
	:	

Revisamos que todo está correcto y creamos la máquina virtual. Aquí podemos ver como el consumo por hora es el mismo que en la calculadora.

Price

1 X Standard B2Is_v2

by Microsoft

[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ

0.0456 USD/hr

Nos descargamos la clave privada SSH.

Download Private Key



 An SSH key pair contains both a public and private key. **Azure doesn't store the private key.** After the SSH key resource is created, you won't be able to download the private key again. [Learn more](#) 

[Download private key and create resource](#)

[Return to create a virtual machine](#)

Y la máquina virtual se empezará a crear y desplegar.

CreateVirtualMachine-vm-practica04-ChristianSalguero | Overview

Deployment

   Delete  Cancel  Redeploy  Download  Refresh

Overview

Inputs

Outputs

Template

*** Deployment is in progress

Deployment name : CreateVirtualMachine-vm-practica04-ChristianSalguero
Subscription : Azure for Students
Resource group : rg-practica04-vm-ChristianSalguero

Start time : 9/22/2026, 5:01:45 PM
Correlation ID : 2a0cf842-a367-4e55-8ed1-b9e31ef47169

Deployment details

Resource	Type	Status	Operation details
 vnetspaincentralTemplateDeployment1790089306624	 Deployment	Created	Operation details
 vm-practica04-ChristianSalguero-nsg	 Network security group	OK	Operation details
 vm-practica04-ChristianSalguero-ip	 Public IP address	OK	Operation details

[Next steps](#)

Máquina virtual creada correctamente.

Deployment

Search

×

«

Overview

Inputs

Outputs

Template

🗑️

Delete

⌛

Cancel

🔄

Redeploy

⬇️

Download

🔄

Refresh

✔️

Your deployment is complete

📄

Deployment name : CreateVirtualMachine-vm-practica04-ChristianSalguero

Subscription : [Azure for Students](#)

Resource group : [rg-practica04-vm-ChristianSalguero](#)

>

Deployment details

✓

Next steps

[Set up auto-shutdown](#)

Recommended

[Monitor VM health, performance, and network dependencies](#)

Recommended

[Run a script inside the virtual machine](#)

Recommended

Go to resource

Create another VM

Scale out your VM

Dato	Valor
Precio por hora mostrado por el portal	0.0456 USD/hr
Precio por hora según tu estimación	0.046 USD/hr
Fecha y hora de creación	17:01:47 22/09/2026

Con la máquina ya creada vuelvo a intentar agregar la orden `Auto ShutDown`.

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vm-practica04-ChristianSalguero

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Connect Start Restart Stop Hibernate Capture Delete Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Auto-shutdown

Essentials

Resource group (move) : rg-practica04-vm-ChristianSalguero

Status : Running

Location : Spain Central

Subscription (move) : Azure for Students

Subscription ID : 2bd3b453-d91d-4da7-a7de-b92c1f20ba30

Tags (edit) : Project : Practica04 Department : Data Environment : Lab Owner : Christian

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name : vm-practica04-ChristianSalguero

Operating system : Linux (ubuntu 24.04)

Save Discard Feedback

Enabled ☒ On ☐ Off

Scheduled shutdown

9:00:00 PM

Time zone

(UTC+01:00) Brussels, Copenhagen, Madrid, Paris

Send notification before auto-shutdown?

☒ Yes ☐ No

Webhook URL ⓘ

Email address ⓘ

christian.salguero@tajamar365.com

Cuando pulsamos en **Save** nos salta este error:

Notifications

×

More events in the activity log →

Dismiss all ▾

!

Updating Schedule

×

Schedule failed to update {"error": {"code": "LocationNotAvailableForResourceType", "message": "The provided location 'spaincentral' is not available for resource type 'Microsoft.DevTestLab/schedules'. List of available regions for the resource type is 'westcentralus,southcentralus,centralus,qatarcentral,australiacentral,austr..."

 Help me troubleshoot

a few seconds ago

✓

Deployment succeeded

×

Deployment 'CreateVirtualMachine-vm-practica04-ChristianSalguero' to resource group 'rg-practica04-vm-ChristianSalguero' was successful.

Go to resource

Go to resource group

7 minutes ago

Que nos indica que la opción no está disponible en la región seleccionada.

Verificar el etiquetado de todos los recursos

Recurso	Tipo	¿Tiene los 5 Tags?	¿Genera coste?
vm-practica04-ChristianSalguero	Virtual machine	Sí	Sí
vm-practica04-ChristianSalguero_OsDisk_1_8fbbd4cdfa48411181b5c94f8945a3fa	Disk	Sí	Sí
vm-practica04-ChristianSalguero-ip	Public IP address	Sí	Sí
vm-practica04-christiansalguero103	Network interface	Sí	No
vm-practica04-ChristianSalguero-nsg	Network security group	Sí	No
vnet-spaincentral-1	Virtual network	No	No

Comprobando cada apartado el único que no tenía las Tags era la **Virtual Network** entonces se los agregué a mano.

Edit tags

×

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags](#)

Tags

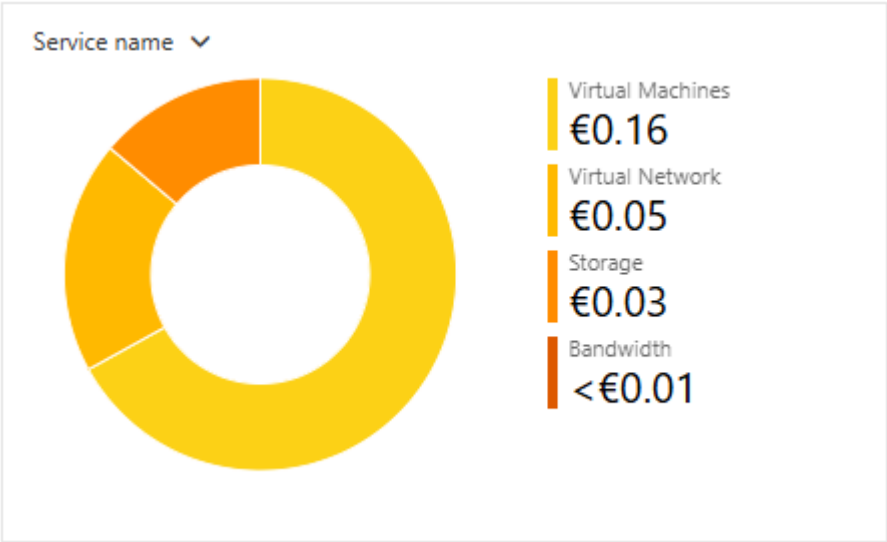
Name ⓘ	Value ⓘ	
Project	: Practica04	
Department	: Data	
Environment	: Lab	
Owner	: ChristianSalguero	
CostCenter	: DataNova-Analytics	
	:	

Resource

vnet-spaincentral-1 (Virtual network)
5 to be added ⓘ

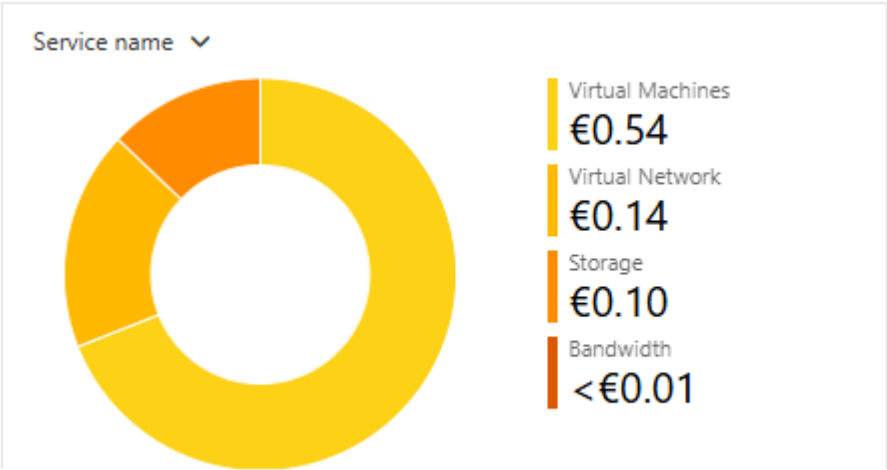
Primera revisión en Cost Analysis

Recurso	Coste acumulado
Máquina virtual	0.16€
Disco	0.03€
IP pública	0.05€
Otros	0.01€
Total (Actual Cost)	0.25€

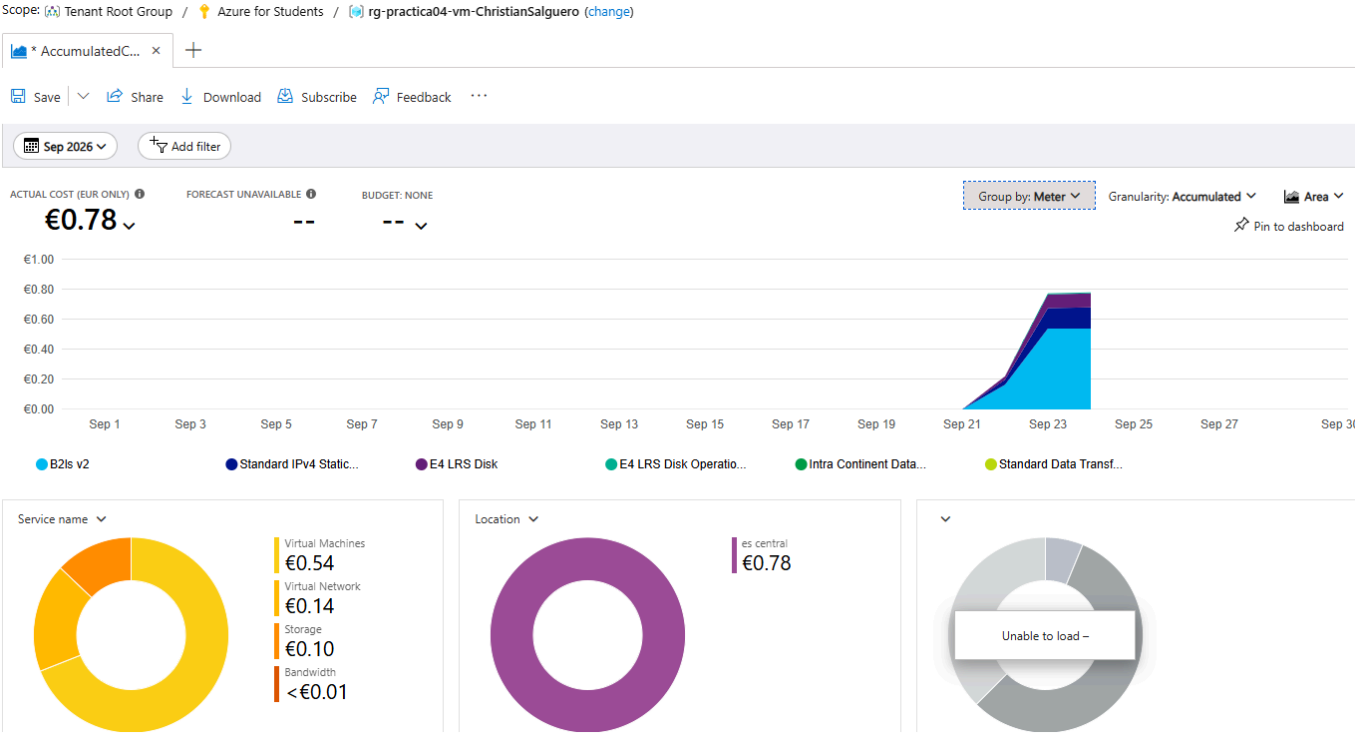


Segunda revisión en Cost Analysis

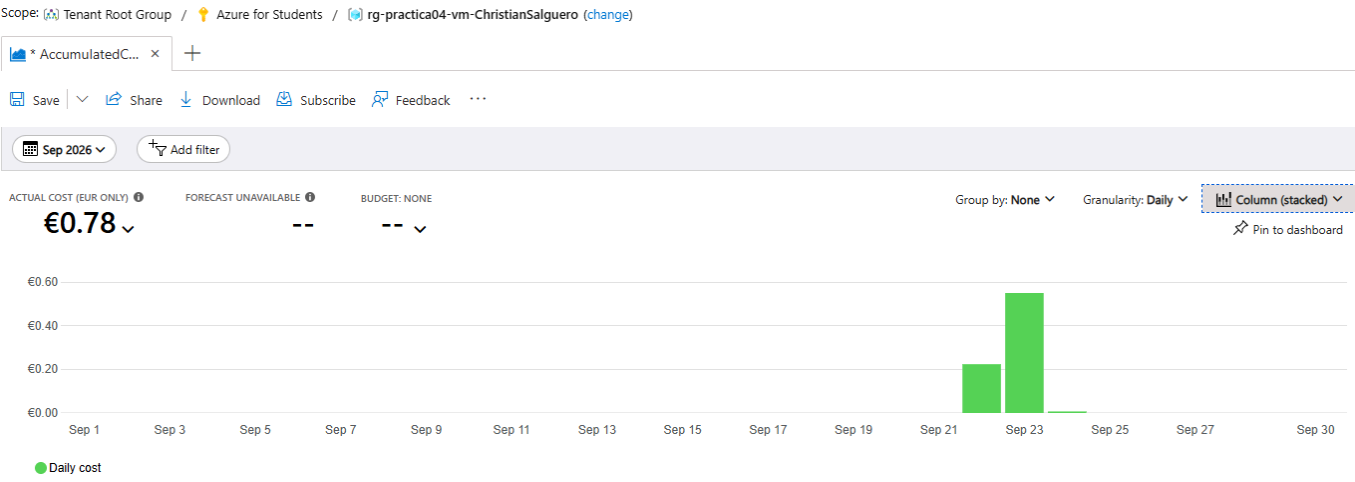
Recurso	Coste acumulado
Máquina virtual	€
Disco	€
IP pública	€
Otros	€
Total (Actual Cost)	€



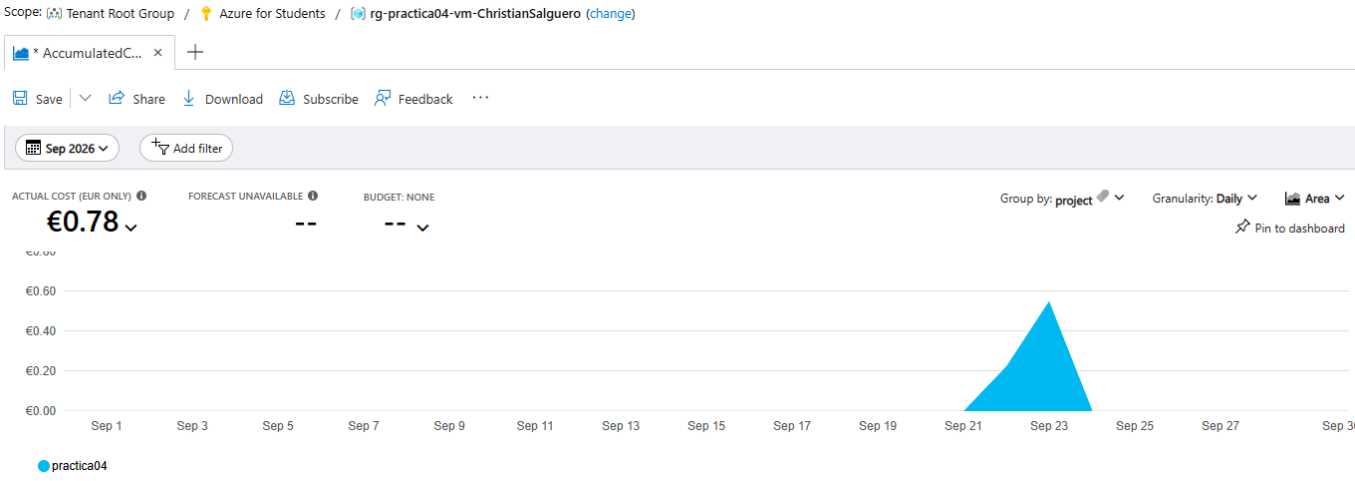
Ver el coste por medidor



Ver el coste diario



Ver el coste por Tag



Consultar la previsión

No se puede visualizar la previsión de forecast porque indica que 3 días es insuficiente para calcularlo.

Dato	Valor
Actual Cost	0.78€
Forecasted Cost (fin de mes)	No disponible
Budget del laboratorio	2€
¿Actual supera el Budget?	No
¿Forecast supera el Budget?	Sí/No

Revisar las alertas

No se ha activado ninguna ya que lo máximo que se ha llegado a gastar ha sido 0.78€ que queda lejos de 1€.

Alerta	¿Se ha activado?	Fecha y hora
Actual 50%	No	n/a
Actual 80%	No	n/a
Actual 100%	No	n/a
Forecasted 100%	No	n/a

Experimento: apagar no es lo mismo que desasignar

Comparar estimación y coste real

Se está comparando con la estimación B. También hay que tener en cuenta que la máquina ha sido apagada por las noches y encendida por el día.

Concepto	Estimado (Sección 3)	Real (Cost Analysis)	Diferencia
Cómputo	1,41€	0,54€	0,87€
Disco	0,20€	0,10€	0,10€
IP pública	0,31€	0,14€	0,17€
Total	1,92€	0,78€	1,14€

Limpieza

Eliminamos el Resource Group.

Home

rg-practica04-vm-ChristianSalguero

Resource group

Search

CreateManage viewDelete resource groupRefreshExport

Overview

Activity log

Access control (IAM)

Tags

Analyze (Preview)

Resource visualizer

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Monitoring

Automation

Help

Essentials

Subscription (move): Azure for Students

Subscription ID: 2bd3b453-d91d-4da7-a7de-b92c1f20ba30

Tags (edit): Project: Practica04Department: DataShow more

ResourcesRecommendations

Filter for any field...

Type equals allLocation equals all

☐ Name ↑

☐ vm-practica04-ChristianSalguero

☐ vm-practica04-ChristianSalguero-ip

☐ vm-practica04-ChristianSalguero-nsg

☐ vm-practica04-christiansalguero103

☐ vm-practica04-ChristianSalguero_key

☐ vm-practica04-ChristianSalguero_OsDisk_1_8fbbd4cdfa48411181b5c94f8945a3fa

☐ vnet-spaincentral-1

Showing 1 - 7 of 7. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

rg-practica04-vm-ChristianSalguero

7 resources being deleted across 1 subscriptions, 2 resource groups

All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
vm-practica04-ChristianSalguero	Virtual machine	Azure for Students	rg-practica04-vm-Ch...
vm-practica04-ChristianSalguero-ip	Public IP address	Azure for Students	rg-practica04-vm-Ch...
vm-practica04-ChristianSalguero-nsg	Network security group	Azure for Students	rg-practica04-vm-Ch...
vm-practica04-christiansalguero103	Network interface	Azure for Students	rg-practica04-vm-Ch...
vm-practica04-ChristianSalguero_key	SSH key	Azure for Students	rg-practica04-vm-Ch...
vm-practica04-ChristianSalguero_OsDisk_1_8fbbd	Disk	Azure for Students	RG-PRACTICA04-VM...
vnet-spaincentral-1	Virtual network	Azure for Students	rg-practica04-vm-Ch...

Enter resource group name to confirm deletion *

rg-practica04-vm-ChristianSalguero

DeleteCancel

Tras esperar unos minutos el grupo de recurso se elimina completamente al igual que todos sus recursos. Como ya no hay recursos con Tags, no puedo filtrar por Tags.

HomeResource Manager

Resource Manager | All resources

Filter resources by regionAre any resources impacted by outagesBrowse all resources in this subscription

Search

CreateManage viewRefreshExport to CSVOpen queryAssign tagsDeleteAdd to service group

Filter for any field...

Subscription equals allResource Group equals allType equals allLocation equals allAdd filter

Group by none

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

No resources match your filters

Try changing or clearing your filters.

CreateClear filters

Learn more

Filter results

Filter

Select options

Search for a filter...

Name

Resource Group

Type

Kind

Location

Edge zone

Tags

No tags available

Informe final

Elemento	Resultado
Nombre de la práctica	Práctica 04 — VM Linux con control de costes
Resource Group	rg-practica04-vm-alumnoXX
Servicios utilizados	Virtual Machine, Managed Disk, Pubilc IP, Virtual Network
Coste estimado antes de desplegar	1,92€

Elemento	Resultado
Budget disponible	2,00€
Alertas configuradas	Actual 50% (1,00 €), Actual 80% (1,60 €), Actual 100% (2,00 €) y Forecasted 100% (2,00 €)
Alertas activadas	Ninguna (el gasto máximo fue de 0,78€)
Coste observado	0,78€
Recurso con mayor coste	Máquina virtual / Cómputo (0,54 €)
Recursos eliminados	Todos los recursos del Resource Group (rg-practica04-vm-alumnoXX) eliminados completamente al finalizar el Día 3
Observaciones	El coste real (0,78 €) fue inferior al estimado (1,92 €) debido a que la Virtual Machine permaneció encendida menos de las 36h previstas. El auto-shutdown no estaba disponible en la región Spain Central, por lo que el apagado lo hice manualmente. Las alertas no se llegaron a disparar y el pronóstico (Forecast) no estuvo disponible por falta de histórico.

Ejercicio final

Un equipo de DataNova Engineering despliega cinco VMs de pruebas con esta configuración:

```
Budget del proyecto           = 50 €
Actual Cost (día 10 del mes)  = 22 €
Forecasted Cost               = 68 €
Apagado automático           = No configurado
Estado de 3 VMs              = Stopped (no deallocated)
```

Responde:

1. ¿Qué alertas se habrán activado con umbrales Actual 50 %, 80 %, 100 % y Forecasted 100 %?

Se ha activado el Forecasted al 100% ya que ahora mismo el Forecasted Cost es mayor que 50€. El resto de alertas no se han activado.

2. ¿Por qué el Forecasted Cost es tan superior al Actual Cost?

Porque el Actual Cost es de 22€ y representa únicamente los gastos acumulados en los primeros 10 días del mes. Azure toma ese ritmo de consumo actual y lo proyecta para todo el mes (el resto de 20 días).

3. ¿Qué parte del gasto de las 3 VMs detenidas se podría eliminar sin borrarlas?

Al cambiar el estado de las VMs de Stopped (apagadas desde su sistema operativo) a Stopped (Deallocated) (desasignadas desde el portal de Azure), Azure libera los recursos hardware y deja de cobrar el cómputo por horas. El disco y la IP pública asignada seguirán cobrándose mientras existan.

4. Ordena estas acciones de mayor a menor impacto en el ahorro inmediato:

- [1] Desasignar las 3 VMs detenidas. (Elimina inmediatamente el cobro por hora del cómputo de 3 máquinas que no se están usando).
- [2] Configurar apagado automático en las 5 VMs. (Reduce masivamente las horas de cómputo activas).
- [3] Cambiar los discos de Premium SSD a Standard SSD. (Ahorro permanente en el almacenamiento).
- [4] Eliminar las IP públicas que no se usan. (Ahorro continuo menor por IP pública reservada).
- [5] Añadir Tags a todos los recursos. (Impacto nulo en el coste directo).

5. ¿Cuál de las acciones anteriores no ahorra dinero pero es imprescindible para analizarlo?

Añadir Tags a todos los recursos.

Aplicar etiquetas no reduce la factura de Azure, pero es un paso indispensable para tener controlado el gasto por proyecto, departamento, etc. Y detectar qué recursos o equipos están generando más gastos.

Referencias oficiales

- Microsoft Learn — Tutorial: Create and manage budgets <https://learn.microsoft.com/azure/cost-management-billing/costs/tutorial-acm-create-budgets>
- Microsoft Learn — Cost Analysis <https://learn.microsoft.com/azure/cost-management-billing/costs/quick-acm-cost-analysis>
- Microsoft Learn — Group and filter options in Cost Analysis and Budgets <https://learn.microsoft.com/azure/cost-management-billing/costs/group-filter>
- Microsoft Learn — Create a Linux virtual machine in the Azure portal <https://learn.microsoft.com/azure/virtual-machines/linux/quick-create-portal>
- Microsoft Learn — States and billing status of Azure Virtual Machines <https://learn.microsoft.com/azure/virtual-machines/states-billing>
- Microsoft Learn — Auto-shutdown a virtual machine <https://learn.microsoft.com/azure/virtual-machines/auto-shutdown-vm>
- Azure Pricing Calculator <https://azure.microsoft.com/pricing/calculator/>